Halward Inspection Services Ltd.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
HIS-2011-0207-DS-4-RA-104	Issued	2011-02-18

UT Thickness Data Sheet - 4-RA-104

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for line 4-RA-104, located within the ATU-100. The scope of work involves the collection of thickness data to support the 5-year Class 1 survey as detailed in HIS-2011-0207.

Kevin Doucette, acting as the lead UT technician, is responsible for the execution of the measurements and the recording of the data. All activities were performed in accordance with the quality and technical requirements of the Hamilton Plant.

Technical Basis and Procedures

Thickness measurements were performed under QSCP-NDT-001 Rev 2, which incorporates requirements for the CUI program, including insulation removal and the use of inspection ports. The field work was further governed by HIS-UT-002 Rev 1, the procedure submitted by Halward Inspection Services Ltd. for acceptance.

The ultrasonic testing was conducted in accordance with the requirements of ASTM E797. All measurements were taken on the A106 Gr.B piping of the L/R exchanger rich side.

Component Design Data

The primary subject, 4-RA-104, has a nominal outer diameter of 114.3 mm and a nominal wall thickness of 8.56 mm. The design pressure is 862 kPa(g) and the design temperature is 127 °C.

The minimum allowable thickness (t-min) for this line is 3.81 mm. This value serves as the retirement thickness for the asset.

Inspection Results and Analysis

Data collected for 4-RA-104 is integrated into the findings of HIS-2011-0207. The analysis focuses on the rich amine service within the ATU-100 to determine current wall loss and corrosion rates.

Based on the 5-year Class 1 survey, the rich amine circuit is currently at half remaining life. This calculation is based on the larger of the long-term or short-term corrosion rates relative to the t-min of 3.81 mm.

Thickness readings, measured 2011-02-18

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	2006-02-24	6.59	6.39	0.049	0.040	>50	2016-02-18	
6-SG-101-02	7.11	3.56	2006-02-24	6.51	6.14	0.061	0.074	34.7	2016-02-18	
6-SG-101-03	7.11	3.56	2006-02-24	6.42	6.17	0.061	0.050	42.9	2016-02-18	
6-SG-101-04	7.11	3.56	2006-02-24	6.28	6.05	0.070	0.046	35.8	2016-02-18	
6-SG-101-05	7.11	3.56	2006-02-24	6.22	5.87	0.085	0.070	27.2	2016-02-18	
6-SG-101-06	7.11	3.56	2006-02-24	6.49	6.27	0.054	0.044	>50	2016-02-18	
6-SG-101-08	7.11	3.56	—	—	6.33	—	—	—	2016-02-18	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	2006-02-24	6.26	5.75	0.093	0.102	21.4	2016-02-18	
6-SG-102-02	7.11	3.56	2006-02-24	6.25	5.80	0.088	0.090	24.8	2016-02-18	
6-SG-102-03	7.11	3.56	2006-02-24	6.26	5.80	0.083	0.092	24.3	2016-02-18	
6-SG-102-04	7.11	3.56	2006-02-24	6.22	5.73	0.084	0.098	22.1	2016-02-18	
6-SG-102-05	7.11	3.56	2006-02-24	6.72	6.46	0.045	0.052	>50	2016-02-18	
6-SG-102-06	7.11	3.56	—	—	6.41	—	—	—	2016-02-18	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2006-02-24	8.27	8.32	0.013	-0.010	>50	2021-02-18	
4-RA-101-02	8.56	3.81	2006-02-24	8.27	8.20	0.023	0.014	>50	2021-02-18	
4-RA-101-03	8.56	3.81	2006-02-24	8.10	7.98	0.042	0.024	>50	2021-02-18	
4-RA-101-04	8.56	3.81	2006-02-24	8.34	8.21	0.028	0.026	>50	2021-02-18	
4-RA-101-05	8.56	3.81	2006-02-24	8.29	8.16	0.029	0.026	>50	2021-02-18	
4-RA-101-07	8.56	3.81	2006-02-24	8.30	8.30	0.020	0.000	>50	2021-02-18	
4-RA-101-08	8.56	3.81	2006-02-24	8.07	8.02	0.044	0.010	>50	2021-02-18	
4-RA-101-09	8.56	3.81	2006-02-24	8.43	8.32	0.017	0.022	>50	2021-02-18	
4-RA-101-10	8.56	3.81	2006-02-24	8.27	8.28	0.026	-0.002	>50	2021-02-18	
4-RA-101-11	8.56	3.81	2006-02-24	8.14	8.04	0.043	0.020	>50	2021-02-18	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2006-02-24	8.41	8.33	0.014	0.016	>50	2021-02-18	
4-RA-102-02	8.56	3.81	2006-02-24	8.15	7.99	0.039	0.032	>50	2021-02-18	
4-RA-102-03	8.56	3.81	2006-02-24	8.27	8.04	0.035	0.046	>50	2021-02-18	
4-RA-102-04	8.56	3.81	2006-02-24	8.24	7.96	0.039	0.056	>50	2021-02-18	
4-RA-102-05	8.56	3.81	2006-02-24	8.19	8.07	0.032	0.024	>50	2021-02-18	
4-RA-102-06	8.56	3.81	2006-02-24	8.15	8.01	0.033	0.028	>50	2021-02-18	
4-RA-102-07	8.56	3.81	2006-02-24	8.29	8.10	0.032	0.038	>50	2021-02-18	
4-RA-102-08	8.56	3.81	2006-02-24	8.05	7.88	0.041	0.034	>50	2021-02-18	
4-RA-102-09	8.56	3.81	2006-02-24	8.18	7.98	0.038	0.040	>50	2021-02-18	
4-RA-102-11	8.56	3.81	—	—	8.19	—	—	—	2021-02-18	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2006-02-24	10.76	10.47	0.035	0.058	>50	2021-02-18	
6-RA-103-02	10.97	4.57	2006-02-24	10.50	10.23	0.058	0.054	>50	2021-02-18	
6-RA-103-03	10.97	4.57	2006-02-24	10.67	10.42	0.038	0.050	>50	2021-02-18	
6-RA-103-04	10.97	4.57	2006-02-24	10.40	9.97	0.068	0.086	>50	2021-02-18	
6-RA-103-05	10.97	4.57	2006-02-24	10.34	10.10	0.057	0.048	>50	2021-02-18	
6-RA-103-06	10.97	4.57	2006-02-24	10.74	10.56	0.030	0.036	>50	2021-02-18	
6-RA-103-07	10.97	4.57	2006-02-24	10.66	10.55	0.029	0.022	>50	2021-02-18	
6-RA-103-08	10.97	4.57	2006-02-24	10.58	10.44	0.035	0.028	>50	2021-02-18	
6-RA-103-09	10.97	4.57	2006-02-24	10.67	10.56	0.030	0.022	>50	2021-02-18	
6-RA-103-10	10.97	4.57	2006-02-24	10.27	10.05	0.065	0.044	>50	2021-02-18	
6-RA-103-13	10.97	4.57	—	—	10.45	—	—	—	2021-02-18	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2006-02-24	8.16	7.95	0.037	0.042	>50	2021-02-18	
4-RA-104-02	8.56	3.81	2006-02-24	8.23	8.14	0.027	0.018	>50	2021-02-18	
4-RA-104-03	8.56	3.81	2010-09-13	8.59	8.56	0.069	0.069	>50	2021-02-18	
4-RA-104-04	8.56	3.81	2010-09-13	8.52	8.60	-0.185	-0.185	>50	2021-02-18	
4-RA-104-05	8.56	3.81	2010-09-13	8.43	8.57	-0.324	-0.324	>50	2021-02-18	
4-RA-104-06	8.56	3.81	2006-02-24	8.09	7.98	0.041	0.022	>50	2021-02-18	
4-RA-104-07	8.56	3.81	2006-02-24	7.97	7.78	0.054	0.038	>50	2021-02-18	
4-RA-104-08	8.56	3.81	—	—	8.18	—	—	—	2021-02-18	

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2011-02-18	_____